

2026-27

CONCEPTS | CLINICALS | MCQS

PEDIATRICS

- Small Patients. Big Questions.
- Growth, milestones & neonatal care
- Vaccines & clinical scenarios
made easy



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PEDIA

GROWTH

Anthropometric parameters-

Weight

Height

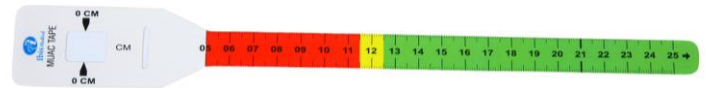
Head circumference

Mid upper arm circumference-shakir tape

Red- malnourished <11.5cm

Yellow - borderline 11.5-12.5cm

Green/normal nutritional > 12.5cm



Skin fold thickness- Harpenden callipers



BMI – weight (kg)/height (m²)

< 5th percentile-undernourished

85-95th - overweight

> 95th - obese

FMGE 22

Weight-

5th month - 2 x

1 year -3 x

2 year -4 x

10 year - 10 x

Height-

Length-1st 2yr of life Measure by Infantometer

At birth-50 cm

1 year- 75 cm

2 year-90 cm

4 - 4.5 year - 100 cm

NEET PG 24,23

Maximum growth take place in- 1st year of life

Length of child increase by 50% in 1st year of life

Height of child double itself in 4 - 4.5 year



Short stature-

Height of child < 3rd percentile or -2SD or Z score of expected

FMGE 25,23

Constitutional delay in growth and puberty	Familial short stature
Temporary delay in skeletal growth and puberty with eventual normal adult height	Genetic short stature due to short parents with normal growth pattern
Often history of delayed puberty in parents	Family members are short but had normal puberty
Delayed onset of puberty	Normal timing of puberty
Bone age < Chronological age	Bone age = Chronological age

Microcephaly-

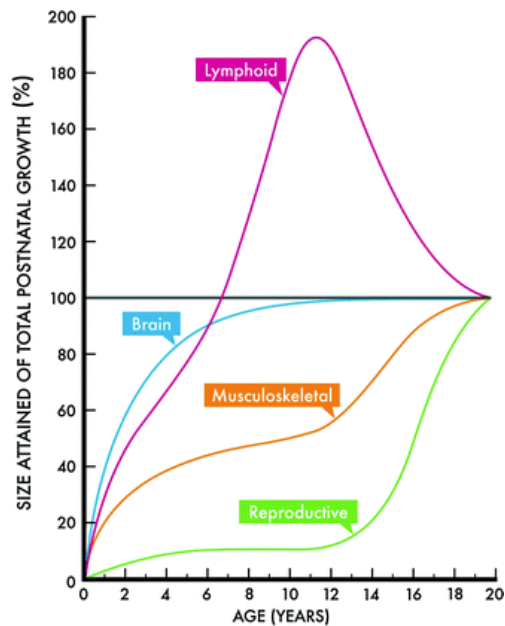
Head circumference < -3Z score of expected according to age and sex of child

Macrocephaly-

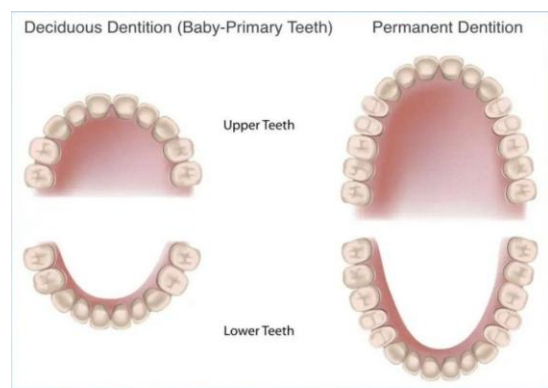
Head-circumference > +2SD of expected according to age and sex of child

Growth and development -

FMGE 23



DENTITION



Dentition	Deciduous	Permanent
Begin	at 6 th month	at 6 year
1 st tooth	Lower central incisor	1 st molar tooth
Last tooth	2nd molar	3rd molar wisdom tooth
Complete by	2.5 to 3 years	12 years except 3 rd molar
Dental formula	ICPM	ICPM

Hutchinson triad- seen in congenital syphilis

1. INTERSTITIAL KERATITIS
2. SNHL
3. HUTCHINSON TEETH

GROSS MOTOR MILESTONES

- 3 months - neck holding
- 6 months - sit with support
- 8 months - sit without support
- 9 months - standing with support
- 12 months - standing without support, Walk with support
- 13 months - walk without support
- 15 months - creep stairs
- 2 year- goes upstairs and come downstairs 2 feet/step
- 3 year- upstairs with alternate feet and downstairs 2 feet/step

FMGE 25,24

FINE MOTOR MILESTONES

FMGE 24

- 3 months - hand regard, Hold and object when place in hand
- 4 months - reaches out to object
- 5 months - Bidextrous grasp
- 6 months - unidextrous grasp
- 7 months - transfer object from 1 hand to another
- 9 months - pincer grasp immature
- 12 months - pincer grasp mature

SOCIAL MILESTONES

FMGE

24

- 2 months - social smile
- 6 months - mirror play
- 7 months - stranger anxiety
- 9 months - wave bye bye

TOWER OF CUBES

FMGE

23

- Number of cubes put on top of each other = Age × 3
- 2 years: Tower of 6 cubes
- 3 years: Tower of 9-10 cubes

- Train without chimney: 2 years
- Train with chimney: 2.5 years
- Bridge: 3 years
- Gate: 4 years
- Steps: 5-6 years

- At 2 years: Copies a straight line
- At 3 years: Copies a circle Handedness gets established
- At 4 years: Copies a rectangle or a cross
- At 5 years: Copies a triangle/ a tilted cross

Developmental quotient-Developmental age/ chronological age X 100

NOCTURNAL ENURESIS

FMGE 25

involuntary urination at night beyond 6 years age

- Mx- Behavioral technique Bed and alarm technique
- Rx- desmopressin
- Imipramine

CAPUT SUCCEDANEUM		vs.	CEPHALHEMATOMA
Feature	Caput Succedaneum		Cephalhematoma
 Definition	Swelling of scalp soft tissue (skin/subcutaneous)		Collection of blood under periosteum (above skull bone)
 Cause	Prolonged/pressured labor (choroid plexus fluid leakage)		Rupture of blood vessels between skull & periosteum
 Location	Crosses suture lines ✓ (can spread over skull)		Does NOT cross suture lines ✗ (confined to one bone)
 Timing of Swelling	Present at birth		May appear hours to days after birth
 Appearance & Feel	Soft, pitting edema (like a boggy cap) ✓ Fluctuant		Firm, well-circumscribed lump (like a subperiosteal blood blister) ✓ Non-fluctuant
 Resolution	Resolves within 2–3 days (may take up to a week)		Resolves over weeks to months (may calcify, need intervention)
 Complications	✓ Benign, no long-term issues		⚠ May cause jaundice, anemia, calcification, ossification



KEY DIFFERENCE

Caput = **Soft** swelling that **CROSSES** sutures & resolves **QUICKLY** ✓

Cephalhematoma = **Firm** lump that **DOES NOT** cross sutures & resolves **SLOWLY** ✗



MNEMONIC: "Caput spreads like a **CAP**; Cephalo stays put like a **HEMATOMA**."

Paladai spoon/katori spoon



BREAST MILK

NEET PG 25, FMGE 25, 23, 22

Start breastfeeding as soon as possible within 1 hour of a childbirth
Exclusive breastfeeding for- 1st 6 months
Amount of milk given to term baby in first 24 hours after delivery - 60ml/kg
Contraindication of breastfeeding- Galactosemia, Lactose intolerance
Breast milk-Lipid - DHA
Vitamin - all except Vitamin D and k
Protein- whey

THERAPEUTIC DOSE OF VITAMIN A

< 6 months - 50000 IU/DOSE

6 to 12 months- 1 LAC

>12 months - 2 lac IU

3 dose are required for therapeutic-

1st at the time of dx

2nd after 24hrs

3rd on 14 day

For prophylaxis 6 dose are required
Start from 9th months with 6 months of interval

FMGE 25,23,22

Acute malnutrition- decrease weight for height or Wasting

Chronic malnutrition- decrease height for age or stunting

Wasting -Weight for height below -2 to -3 Z score Or 70 to 79% of expected

Stunting- Height for age below -2 to -3 Z score or 85-89% of expected

Primitive neonatal reflexes-

Present at birth

Rooting reflex- 32 week of gestation



Rooting Reflex

Asymmetric tonic neck reflex- appear at 35 week of gestation and disappear at 3 to 5 month



Moro reflex and Palmar reflex - 28 week gestation and disappear at 3 month of postnatal age



Reflexes Appear after birth-

FMGE 24

Symmetric tonic neck reflex- Appear at 4 to 6 month Disappear at 8 to 12 months

Parachute reflex- appear 8 to 9 months and never disappear



FLUID AND ELECTROLYTES CALCULATION

FMGE 23

Calculation of 24 hours

Maintenance of fluid required in children

1st 10kg - 100 ml/kg

Next 10kg - 50 ml/kg

Weight beyond 20 kg - 20 ml/kg

Mx of shock in children

IV fluid bolus → normal saline

20 ml/kg □ upto 3 times--Inotropes □ dopamine- □ catecholamine resistant shock □ iv hydrocortisone

FLUID OF CHOICE for shock in severe acute malnutrition - ringer lactate with 5% dextrose

NEONATAL RESUSCITATION

FMGE

25,23

Suction during neonatal resuscitation- mouth f/b nose pressure 80 to 100 mmHg

Indication of positive pressure ventilation -

Heart < 100

Apnea

Gasping

Device used for PPV- self inflating bag

C.I. of bag and mask ventilation- CDH

Most sensitive indicator of effective positive pressure ventilation- Improvement in HR

Use of O₂ during neonatal resuscitation- If chest compression is required

Chest compression indication - Heart < 60/min despite adequate PPV

Ratio of CC: PPV- 3:1

Use O₂- 100%

Injection adrenaline-

-If heart < 60/min despite CC and PPV

-0.01mg/kg/dose can be given upto 3 times

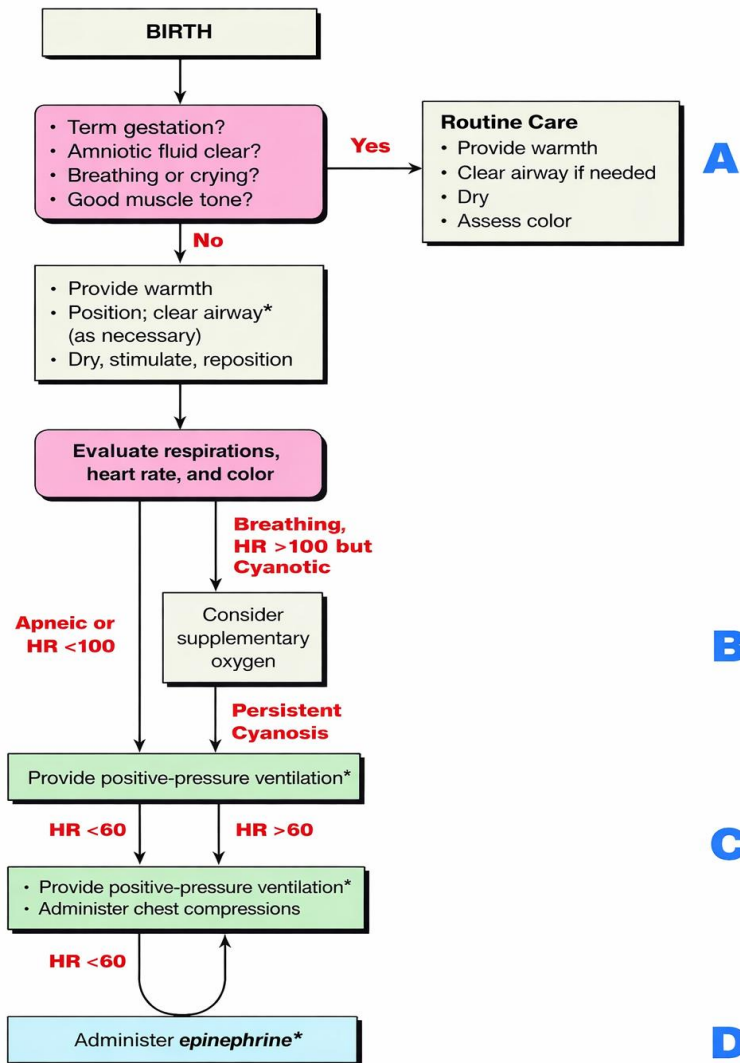
-0.1ml/kg of 10000 adrenaline

Route IV

Resuscitation of baby born through meconium stained liquor- Previously check if baby is

-Vigorous-- heart > 100, good muscle tone, strong respiratory efforts

Non vigorous- ET intubation, Tracheal suction is done



* Tracheal intubation may be considered at several stages

FEEDING OF PRETERM NEONATES

Gestational age	Recommend initial mode of feeding
< 28 week	IV fluids +/- TPN
28-31 week	Orogastric tube 
32-34 week	Katori spoon or Paladai feeding
>34 week	Breast feeding

NEONATAL HYPOTHERMIA

Axillary temperature $<35.5^{\circ}\text{C}$

Classification

Cold stress - $36-36.4^{\circ}\text{C}$

Moderate hypothermia - $32-35.9^{\circ}\text{C}$

Severe hypothermia - $<32^{\circ}\text{C}$

Devices keep neonates warm

Radiant warmer	Incubator
 Heating element canopy Heater Mattress Radiation keep baby warm	 Convection keep baby warm

Delayed cord clamping?

- For at least 60 seconds
- Help to prevent – Anemia, BT, IVH, NEC (doesn't decrease mortality rate)
- Late benefits – prevent iron deficiency anemia by 3-6 months of life

- Negative points:
 - Increase risk of jaundice
 - Asymptomatic Polycythemia ($>22\text{g/dl}$)
- Delayed fall will indicate Leukocyte adhesion defect (LAD)

Kangaroo mother care (KMC)

- o Method of care of preterm or LBW
- o Skin to skin (STS) contact
- o Exclusive breast feeding
- o Early discharge

Components of KMC:

- o Semi reclining position (To prevent gastric reflux)
- Kangaroo positioning -Skin to skin contact between mother & neonate in vertical position between mother breasts under her clothes
- o Monitoring
 - o Feeding
- o Duration – do till the neonates tolerates the position
- o Indicates by sweating or refusal to stay in KMC position.

Best weight to start KMC is

- 1800 grams or <1800 grams but hemodynamically stable
- Stop KMC when baby reaches term gestation or 2.5 kg
- Providing KMC for 1-2 hour / sitting

Discharge criteria:– Gaining weight at least 15-20 gram per kg per day for 3 consecutive days

Pneumonia

In neonates < 3 weeks of age:

Bacteria is the most common cause of pneumonia.

B. Streptococcus (Streptococcus agalactiae).

The second most common cause is E. coli.

In India, the most common cause is Klebsiella > E. coli.

In children 3 weeks to 4 years of age: viral > bacterial causes.

India/World: RSV is the most common in the first two years of the child, followed by rhinovirus.

India/World: The most common bacterial cause is Streptococcus pneumoniae > H. influenzae.

In children of five years or older, the most common cause is Mycoplasma pneumoniae > Streptococcus pneumoniae > Chlamydia pneumoniae.

Clinical Features

Tachypnea, or rapid breathing, is the most consistent feature clinically seen in childhood pneumonia.

It has an age-dependent respiratory rate.

If the age of the child is < 2 months, the respiratory rate is ≥ 60 /minute.

If the age of the child is 1-5 years: respiratory rate: ≥ 40 /minute.

If the age of the child is 2–12 months, respiratory rate: ≥ 50 /minute.

In infants with pneumonia:

Tachypnea

Fever (moderate-grade fever—viral cause and high-grade—bacterial cause)

Cough, respiratory distress, irritability, poor feeding.

GIT symptoms (20-25%): diarrhea, vomiting, abdominal distension.

Danger Signs:

Severe protein energy malnutrition.

Persistent vomiting: >3 vomiting/hour.

Lethargic with abnormally sleepy or difficult to awaken.

Convulsions or seizures.

Inability to feed and drink.

Stridor at rest/Stridor in a calm child.

Treatment

A. Empirical therapy

Pneumonia: Treated on an OPD basis.

Drug of choice: oral amoxicillin in 80 mg/kg/day in 2 divided doses x 5 days.

Reviewed in 48 hours.+ SUPPORTIVE THERAPY continued.

If complications are seen, then switch to parenteral antibiotics and advise higher center referral.

Severe pneumonia

According to IMNCI, in Primary Health Centre.

The first dose of antibiotic should be given and then referred to a higher centre.

Hospital admission

Parenteral antibiotics started—a combination of ampicillin + gentamycin.

Parenteral antibiotics continued for 10 days.

NEONATAL HYPOGLYCEMIA

FMGE 25

Blood glucose <40 mg/dl or plasma glucose <45mg/dl

Most common complain - tremors

Trt -

If symptomatic → IV 10% dextrose 2ml/kg

PERINATAL ASPHYXIA

Failure to initiate and sustain breathing at birth leading to hypoxia, hypercapnia, and acidosis

Part of brain most commonly affected due to asphyxia

Term neonate	Preterm neonate
Parasagittal area → spastic quadriplegia	Periventricular area - spastic diplegia

Diagnosis

APGAR score - 0-3 >5 min

Cord blood pH <7

CNS dysfunction

SCORE	0 points	1 point	2 points
A ppearance (Skin color)	Cyanotic / Pale all over	Peripheral cyanosis only	Pink
P ulse (Heart rate)	0	<100	100-140
G rimace (Reflex irritability)	No response to stimulation	Grimace or weak cry when stimulated	Cry when stimulated
A ctivity (Tone)	Floppy	Some flexion	Well flexed and resisting extension
R espiration	Apneic	Slow, irregular breathing	Strong cry

Total score - 10

Minimum score - 0

APGAR score >7 normal

APGAR score 0-3 → severely depressed neonate

STAGING OF HYPOXIC ISCHEMIC ENCEPHALOPATHY

Parameters	Stage I	Stage II	Stage III
Level of consciousness	Hyperalert	Lethargic / Hypoactive	Coma / Unresponsive
Muscle Tone & Activity	Normal	Hypotonia	Flaccid
Reflexes	Exaggerated Moro & stretch reflexes	Weak suck & Moro reflex	Absent reflexes
Seizures	Rare or absent	Frequent seizures	Frequent → not severe

RESPIRATORY DISTRESS SYNDROME

NEET PG 23

Most common cause of respiratory distress is preterm neonate

Deficiency of mature surfactant

Most important content of surfactant - DPPC/Lecithin

Trt

Mild - CPAP +/- O₂

Moderate to severe - intratracheal surfactant (INSURE technique)



Prevention of RDS/hyaline membrane ds

Antenatal steroids-

Dexamethasone - 6mg, 4 dose, 12hours apart

Betamethasone - 12mg, 2 dose, 24 hours apart

MECONIUM ASPIRATION SYNDROME

Respiratory distress occurring soon after birth in an infant born through meconium-stained amniotic fluid (MSAF), with evidence of meconium in the trachea/lungs

Chest Xray findings -

Hyperinflated lungs

Segmental collapse



TRANSIENT TACHYPNEA OF NEWBORN

Benign, self-limited respiratory distress due to delayed clearance of fetal lung fluid in the immediate newborn period

Most common seen in babies born through C section

Chest xray findings -

Fluid in interlobar fissure

Prominent broncho vascular markings



NEONATAL JAUNDICE

Physiological jaundice	Pathological jaundice
After 24 hours of life	Within 24 hours of life
Unconjugated (indirect) bilirubin	Usually unconjugated, but may be conjugated
Resolves by 7 days	Persists > 14 days (term)

Breastfeed jaundice	Breastmilk jaundice
Due to inadequate breastfeeding	Due to pregnanediol and free fatty acids ↓ Interfere with conjugation of bilirubin

Trt -

First line - Phototherapy

Severe cases - Exchange transfusion

DRUG OF CHOICE

PDA in preterm neonate-ibuprofen/indomethacin
Keeping Ductus arteriosus patent- Alprostadil
Side effect will be Apnea

ACUTE DIARRHEA

MCC Rotavirus

Mc finding- isotonic dehydration
Metabolic acidosis,Hypokalemia

Mx of acute diarrhea in children

If no dehydration- give only WHO ORS 5 TO 10 ML/KG/LOOSE STOOLS

Some dehydration -75ml/kg next 4 hours WHO ORS or IV RL in 5% dextrose

Severe dehydration - <1 yr → 100ml/kg - 1st 30ml in 1 hour
Next 70ml in 5 hours total 6 hours

> 1yr→
1/2hr 30ml
2.5hr 70ml total 3 hours

IV fluid of choice with diarrhea and dehydration- RL+ 5% dextrose >> RL

NEW FORMULA WHO-ORS

CONTENT

NaCl : 2.6g

KCl: 1.5g

Trisodium.citrate:2.9g

Glucose: 13.58

CONCENTRATIONS

Na+ - 75mM

K+ -20mM

CL -65mM

Citrate- 10mM

Glucose-75Mm

Water: IL

Total osmolarity 245 mOsm/L

Approach to child with cough/ difficulty in breathing-

FMGE 25,22

Fast breathing	Chest indrawing	General danger signs	category	Rx
–	–	–	No pneumonia Only cough and cold	Home care advice
–/+	–/+	–	pneumonia	Oral antibiotics+ home care
–/+	–/+	+	Severe pneumonia	1st dose antibiotic and refer to higher center for further treatment

Tetralogy of fallot- boot shape heart

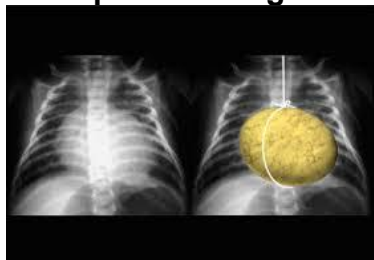
FMGE 25



TAPVC- snowman appearance

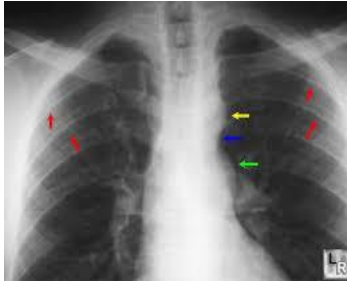


Transposition of great arteries- lemon on string



Coarctation of aorta- figure of 3

FMGE 24



FMGE 23,22

Kwashiorkor	Marasmus
Protein deficiency	Energy protein malnutrition Calories deficit
Loss of appetite and stunted growth in children	Shriveled appearance of child
Building eyes, swollen face	Prominent ribs, wrinkled skin
Stomach protrudes out	Retarded physical and mental growth
Can be cured by giving them sufficient amount of protein rich diet	Can be cure by giving sufficient amount of protein rich diet

CRETINISM (congenital hypothyroidism)

FMGE 24

Most common treatable/preventable cause of mental retardation /intellectual disability in children

c/f-

child with dry and rough skin, big tongue, rough hair

Dx - diagnose after 48-72 hrs of delivery

Trt- Oral levothyroxine

Benign Birthmarks

1)Eyelid Patch (Nevus Simplex)



Faint pink/red patch on eyelid
 Due to capillary dilation
 Fades within 1 year
 Commonly called "angel kiss"

2) Salmon patch



Seen on nape of neck / forehead
 Also called "stork bite"
 Blanchable pink patch
 Usually fades with age (neck may persist)

3) Café-au-lait Spots



Light brown macules
 Smooth borders ("coast of California")
 >6 spots → Think Neurofibromatosis (NF1)
 May be present at birth or appear later

4) Congenital Dermal Melanocytosis (Mongolian Spot)



Bluish-gray patch (usually sacral area)
Common in darker-skinned infants
Can be mistaken for bruise
Fades in childhood

5) Epstein pearls



- Epithelial inclusion cyst mainly on hard palate
- Due to accumulation of epithelial cells
- Also seen over tip of penis

6) Harlequin baby



- A.k.a collodion baby, Ichthyosis vulgaris
- AR condition
- Its lipid membrane barrier defect

NEONATAL SEIZURE

22

MC type Subtle seizure
MCC - Hypoxia

NEET PG 25, FMGE

Preferred CNS imaging modality for neonates- Transcranial USG
Doc for neonatal seizure- Phenobarbitone > Levetiracetam

Most common cause of seizure in <5 year of children - febrile seizure

Management of febrile seizure- buccal midazolam

Management of status epilepticus- seizure lasting >5 minutes without gaining consciousness.

Maintain airway, breathing, circulation



IV access/ nasal or buccal midazolam



Injection of lorazepam/midazolam



Injection phenytoin 20mg/kg

DOC for absence seizures- Ethosuximide
Drug for west syndrome- inj. ACTH

Alagille syndrome

Hemivertebrae/Butterfly vertebrae
Conjugated hyperbilirubinemia
Triangular facies
Pulmonary stenosis
Eye abnormalities

Rett Syndrome

X linked dominant inheritance
M/c in: Females
M/c gene affected: MECP2 gene
Head circumference at birth: Normal
C/f:
Gradual developmental delay
Acquired microcephaly
Loss of purposeful hand movements
Stereotypic hand wringing movements
FDA approved drug: Trofinetide